STaR-Body: Identity-Preserving Evidence Routing for Spatiotemporal Human Mesh Refinement

Anonymous CVPR submission

Abstract

Video human mesh refiners routinely condition on 2D keypoints, yet the interface through which this evidence reaches the body is rarely isolated: keypoints are pooled or concatenated into one vector that is broadcast to every joint. We ask whether preserving the identity of each landmark and routing it to the joints it constrains matters once the evidence is real. An audit of our own joint-indexed feedback motivates the question: its landmark observations came from a head that never received gradient, so it routed drift from a frozen reference model rather than image evidence. STaR-Body therefore turns external detector keypoints into identity-bearing tokens that carry the residual to the current body projection, the projection itself and the detector confidence. Twenty-four SMPL joint queries read these tokens through attention with a kinematic-tree distance bias; the feedback is recomputed at every refinement iteration, enters before part pooling and masked temporal exchange, and starts from a zero-initialised output that preserves the inherited prediction. On 3DPW, all variants start from one parent, share a 1,000-step budget and three seeds, and are tested once through validation locks; we compare routing with no feedback, capacity-matched pooling, ordered-vector conditioning, identity-free sets, unbiased and wrongly biased routing, and a weaker detector, using paired hierarchical-bootstrap intervals. ***[TODO: One sentence with the main test result, e.g. "Routing lowers test MPJPE by TBD mm relative to ordered-vector conditioning (95% CI TBD)"; if the CIs include zero, state that identity-preserving routing performs on par with global conditioning and summarise what does matter.]***

1. Introduction

Monocular human mesh recovery estimates an articulated SMPL body [20] from images. Single-image regressors [10, 12, 16] now provide strong initial estimates, and video methods refine them with temporal context [5, 14, 23] and with detected 2D keypoints [24, 25]. Keypoints are an attractive refinement signal: they are cheap to obtain from mature detectors [4, 34], they are expressed in the same image plane as the body projection, and each of them constrains a specific part of the kinematic tree. How this evidence should *enter* a refiner, however, is rarely studied in isolation. Common interfaces pool landmark residuals into a single vector or concatenate the keypoints into an ordered per-frame vector; either way, a single per-frame feature computed by a global network conditions every joint. Pooling discards which landmark produced a residual, and concatenation keeps identity only by position, leaving a global network to discover which joint each coordinate should move.

This paper asks a narrow question: *does preserving landmark identity, and routing each landmark anatomically to the joints it constrains, matter once the evidence is real?* We were led to it by an audit of our own earlier refiner. An earlier variant of our refiner routed 17 landmark tokens to 24 joint queries and reported small single-seed validation gains over a mean-feedback control (PVE 72.82 *vs*. 73.05 mm). Its landmark observations, however, came from an observation head whose 2D losses were zero in every training stage; the head never left its initialisation and its output equals the projection of the frozen reference model (Sec. 3.2; **[TODO: confirm]**). That refiner therefore routed the drift of the current body from a frozen reference, not image evidence, and its mean control had 206 times fewer parameters. The earlier comparison cannot answer the question above.

We therefore rebuild the feedback interface around external detector keypoints and design a study that changes one factor at a time. STaR-Body encodes each ViTPose [34, 35] landmark as a token that carries its residual to the current body projection, the projection itself, its detector confidence and a learned identity embedding. Twenty-four SMPL joint queries read these tokens through multi-head attention with a soft kinematic-tree distance bias. The routed feedback is recomputed at every refinement iteration as the body moves, enters each joint before anatomical part pooling and masked temporal exchange, and is zero at initialisation, so every variant starts from exactly the inherited prediction (Fig. 1). Around this router we place controls that isolate the evidence source (none, self-referential, external), the interface (capacity-matched pooling, ordered concatenation, identity-free set attention, routing), the anatomical prior

(none, correct, deranged) and the evidence quality (ViTPose *vs*. the OpenPose [4] detections shipped with 3DPW [29]). All variants start from one parent checkpoint, share a 1,000-step budget and a pre-registered evaluation step, are trained with three seeds, and are evaluated on the 3DPW test split exactly once through immutable validation locks.

Our contributions are (i) a diagnostic showing that joint-indexed feedback built on an untrained observation head routes reference drift, which explains why an apparent interface effect can arise without image evidence; (ii) an identity-preserving evidence router that turns external keypoints into anatomically routed, per-iteration joint feedback inside a spatiotemporal refiner, with bounded rotation updates and a start that reproduces the inherited prediction exactly; and (iii) a controlled, multi-seed study with paired sequence-clustered statistics that separates evidence, identity, routing, anatomical prior and evidence quality. **[TODO: Close the paragraph with the answer once test numbers exist, *e.g*. "Routing real evidence improves over ordered-vector conditioning and pooling on the 3DPW test split, whereas routing self-referential evidence does not"; if routing is not distinguishable from global conditioning, say so and state the finding that does hold (*e.g*. evidence source matters more than interface).]**

2. Related Work

Regression and geometric feedback. HMR [12], SPIN [16] and HMR 2.0 [10] established direct SMPL regression; CLIFF [19] and CameraHMR [21] add full-frame and perspective cues, while BEDLAM [3] shows the value of synthetic training data. PARE [15] attends to body parts, and PyMAF [39], PyMAF-X [40] and ReFit [30] iteratively resample image features at the current mesh or markers. TokenHMR [7], MEGA [9] and ScoreHMR [26] change the pose representation or the inference procedure, PromptHMR [32] conditions on spatial and semantic prompts, and HSMR [33], NLF [22] and Multi-HMR [2] broaden the body model, the output space or the number of people. ReFit already feeds per-keypoint reprojection signals into recurrent updates, and HybrIK [18] maps keypoints to joint rotations analytically; landmark identity is thus not new in itself. These works decide *where* evidence is sampled or how it is solved for; we isolate, under matched controls, the interface through which already-localised 2D evidence is assigned to joints inside a learned refiner.

Video and keypoint-conditioned reconstruction. VIBE [14], TCMR [5] and GLoT [23] aggregate per-frame image features over time. WHAM [25] and GVHMR [24] additionally consume detected 2D keypoint sequences, which enter the network as ordered per-frame vectors together with image features; TRAM [31], GLAMR [37], HumanMM [42] and OnlineHMR [43] address camera motion, multi-shot input and streaming. Our ordered-concatenation control follows this conditioning style inside our refiner, so that it differs from routing only in the interface. We do not model world-grounded trajectories and evaluate camera-space bodies only.

Sets, tokens and permutation invariance. Deep Sets [38] characterise permutation-invariant functions as pooled per-element embeddings, and the Set Transformer [17] pools sets with learned queries in attention [28], the token mechanism that also underlies vision transformers [6]. A set of keypoint residuals with no identity is permutation invariant, whereas anatomy is not. Our pooled and identity-free set controls instantiate these two families, and the router adds learned landmark identities and a kinematic prior to the Set Transformer's query-based pooling. The shortest-path attention bias follows the spatial encoding of Graphormer [36], and the zero-initialised output follows ReZero [1] and ControlNet [41]; we claim neither as a contribution.

3. Method

3.1. Setting and inherited model

Given an offline monocular person track, STaR-Body predicts for every target frame t a camera-space SMPL state $x_t = (R_t, \beta_t, c_t)$ with 24 joint rotations, ten shape coefficients and a crop camera $c_t = (s_t, t_{x,t}, t_{y,t})$. All variants in this paper share one trained *parent*, which we keep fixed and describe only briefly. Each eight-frame target window reads 192 HMR 2.0a tokens per frame [10] and a 17-frame Wan/VACE context [11, 27] compressed to 30 memory tokens; both encoders are frozen, and the video-generation models Wan and VACE serve only as feature extractors. Twenty-six body queries Z (24 pose, shape, camera) are decoded by blocks that mix a spatial-first order $Z_{\mathrm{ST}} = A_T(A_S(Z))$ and a temporal-first order $Z_{\mathrm{TS}} = A_S(A_T(Z))$ of attention,

$$Z' = a_1 Z_{\mathrm{ST}} + a_2 Z_{\mathrm{TS}}, \qquad a = \mathrm{softmax}\left(g([Z_{\mathrm{ST}}; Z_{\mathrm{TS}}])\right), \tag{1}$$

over three weight-shared iterations that predict a residual to a frozen reference body, using the continuous 6D rotation representation [44]. A track-conditioned residual, bounded componentwise by 0.3, maps a fixed prediction-and-image prototype of the track to a shape correction shared by all of its frames. The final stage, a part-aware recurrent rotation refiner (Sec. 3.5), is where landmark evidence enters; only its feedback module differs between variants.

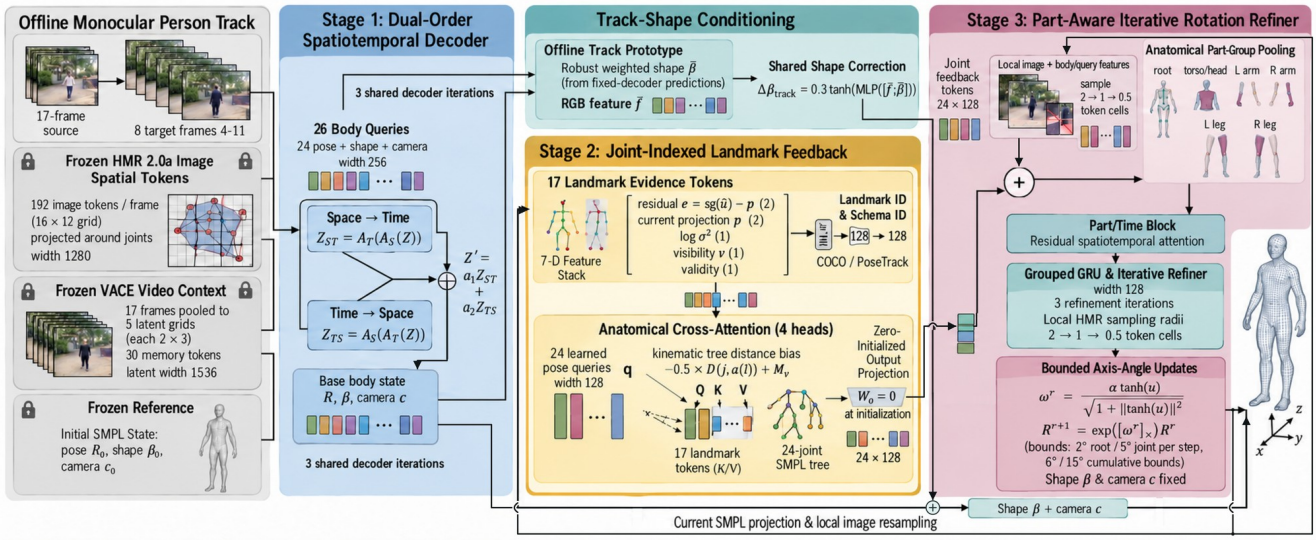


Figure 1. **STaR-Body overview.** A frozen dual-order spatiotemporal decoder reads HMR 2.0a image tokens and Wan/VACE video context and predicts a residual to a frozen reference body; a track-conditioned residual shares shape over a person track. The final stage refines rotations for three iterations: landmark evidence is routed to 24 joint queries, added before part pooling and masked temporal exchange, and converted into bounded axis-angle updates; projections and local image samples are recomputed at every iteration. **[TODO: Redraw stage 2 for C11: evidence tokens come from an external detector (residual $u-p$, projection p, $-2\log c$, c, validity) instead of the inherited head; the drawn step bounds $2^\circ/5^\circ$ are the scales α, the tight cumulative bounds are $5.2^\circ/13.0^\circ$ (Sec. 3.5).]**

3.2. What did joint-indexed feedback route?

This earlier variant, denoted C10, read 17 landmark observations $\hat{u}_{tl}$ from an observation head of the parent, together with a log-variance and a visibility logit, and formed the residual $\mathrm{sg}(\hat{u}_{tl}) - p_{tl}(x^r)$ to the current projection $p_{tl}(x^r)$. Every 2D loss weight was zero in all stages that produced the parent and C10, so the head never received gradient. Its coordinate layers remain at their zero initialisation, which makes $\hat{u}_{tl}$ the projection of the frozen reference body; the log-variance stays at its bias of -4 and the visibility at $\sigma(0) = 0.5$. The routed residual is therefore $p_{tl}(x^{\mathrm{ref}}) - p_{tl}(x^r)$: the image-plane drift of the current body from the reference, with constant uncertainty features. **[TODO: confirm on the archived parent, C00 and C10 checkpoints with scripts/inspect_checkpoint_v43.py; report the measured verdict (all coordinate weights exactly zero, log-variance bias -4) here and in Supp. S5.]** Such a signal can still help a refiner, for instance as a learned pull towards the reference, but it contains no image evidence, so an advantage of routing it cannot be attributed to evidence assignment. Two further limitations affect the archived comparison: the mean-feedback control (C00) mapped four averaged scalars through 512 parameters, against 105,472 for the router, and both runs used a single seed with validation-only evaluation. Relative to its parent, archived C10 changed validation MPJPE by -0.07 mm with a sequence-bootstrap 95% interval of $[-0.21, +0.03]$ (Supp. S5). We keep this self-referential router as variant A2 and its mean control as A1, and build the method below on external evidence.

3.3. Identity-bearing evidence tokens

An external top-down detector, ViTPose [34, 35] (ViTPose++-H, COCO-17 expert), runs on the same 256×256 crop as the image backbone and returns, for each landmark $l \in \{1, \ldots, 17\}$, a position u_{tl} in normalised crop coordinates and a confidence $c_{tl} \in [0, 1]$. Let $p_{tl}(x^r)$ be the COCO landmark regressed from the current SMPL mesh at refinement iteration r and projected with the crop camera. Each landmark becomes a seven-dimensional feature

$$\phi^r_{tl} = \left[\, u_{tl} - p_{tl}(x^r);\ p_{tl}(x^r);\ -2\log c_{tl};\ c_{tl};\ v_{tl} \,\right], \quad (2)$$

where v_{tl} marks finite keypoints with positive confidence in valid frames, invalid entries are zeroed, and residual, projection and log-confidence are clipped to $[-2, 2]$, $[-4, 4]$ and $[-6, 6]$. The term $-2\log c$ plays the role of a log-variance and vanishes for confident detections. A two-layer MLP maps ϕ^r_{tl} to width 128, and a learned landmark embedding e_l plus a schema embedding are added, so that the token z^r_{tl} carries its anatomical identity independently of its content. Because $p_{tl}(x^r)$ is recomputed from the current body, the residual shrinks or changes direction as the refiner moves the joints; the evidence is a closed-loop signal rather than a fixed input.

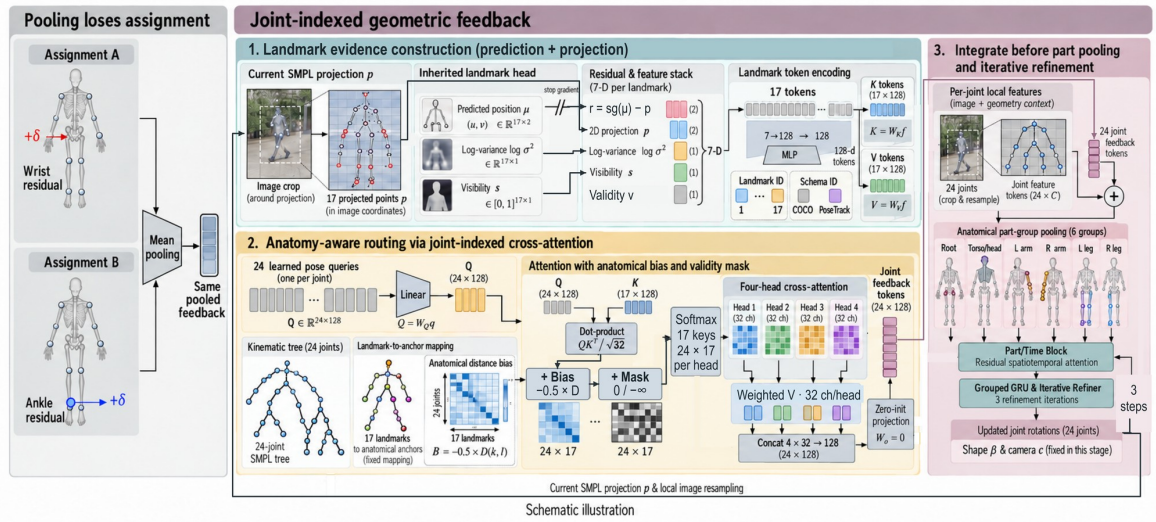


Figure 2. **Identity-preserving routing** (schematic). Left: exchanging a wrist and an ankle residual leaves a mean-pooled feedback vector unchanged. Right: every landmark becomes a token with identity and schema embeddings; 24 joint queries read the 17 tokens through four-head attention with the kinematic-tree bias $-D/2$ and a validity mask, and the zero-initialised output is added to each joint before part pooling and the recurrent update. **[TODO: Redraw panel 1 for C11: the evidence comes from an external detector (ViTPose keypoints u and confidences c) instead of the inherited landmark head.]**

3.4. Anatomically biased routing

Twenty-four learned joint queries q_j read the evidence tokens of a person-frame with four-head attention (Fig. 2),

$$A^h_{jl} = \mathrm{softmax}_l\Big(\frac{(W^h_Q q_j)^\top W^h_K z^r_{tl}}{\sqrt{d_h}} - \frac{D(j, a(l))}{2} + M_{tl}\Big), \tag{3}$$

with $d_h = 32$, where $a(l)$ is the anatomical anchor of landmark l (the head joint for facial landmarks and the corresponding SMPL joint for limb landmarks), D the shortest-path distance on the SMPL kinematic tree and $M_{tl} \in \{0, -\infty\}$ the validity mask. The bias favours nearby anatomy without forbidding long-range reads, *e.g.* a wrist query may still consult the shoulder. The attention-weighted values of the four heads are concatenated, projected, and mapped by a final matrix W_o to 24 joint feedback vectors $f^r_{tj} \in \mathbb{R}^{128}$. We initialise $W_o = 0$, so the feedback is exactly zero at the start of training while W_o receives gradient immediately; together with zero-initialised residual blocks, every variant reproduces the parent's predictions before its first update. A frame without any valid landmark receives zero feedback. The router trains 105,472 parameters, as many as C10. For fusion with the inherited head (B6), the 17 head tokens and the 17 external tokens, each with their own identity embeddings and anchors, form one set of 34 tokens.

Controls as single-factor changes. Every control keeps the token features of Eq. (2) and the zero-initialised output, and changes one factor (Tabs. 2 and 4). The *pooled* control (B1) removes identity and routing: the per-token MLP outputs are averaged over valid landmarks, as in Deep Sets [38], passed through a 128–256–128 MLP and broadcast to all joints (100,096 parameters). The *ordered-vector* control (B7) keeps identity by position: the 17 encoded tokens are concatenated in slot order, mapped by a linear layer and an MLP, and broadcast to all joints, mirroring how keypoints condition WHAM-style models (329,344 parameters, 3.1× the router). The *set* control (B2) keeps the joint queries but removes identity embeddings and bias, which yields Set-Transformer-style pooling by attention [17] that is invariant to permuting landmarks. B3 keeps identity but removes the bias, and B5 keeps identity with a bias whose landmark columns are permuted by a fixed derangement, *i.e.* a wrong anatomy with identical bias statistics. Note that the projected coordinates p_{tl} may reveal identity implicitly even when embeddings are removed; B1 and B2 therefore test whether *explicit* identity and routing help at this budget, not whether identity is recoverable in principle.

3.5. Part-aware iterative rotation refinement

At iteration r, each joint feature combines its contextual pose query, the current body state and local HMR 2.0a samples around its projection, whose radius shrinks from two to one to half a token cell. The routed feedback f^r_{tj} is added to this per-joint feature *before* the joints are pooled into six anatomical groups (root, torso/head, two arms, two legs). Two residual blocks then exchange information across groups within a frame and across time within a group; temporal attention is masked by source-frame identi-

ties so that duplicated or invalid frames and track gaps never act as independent evidence. The group context is broadcast back to the joints, and a group-specific GRU, whose state persists over the three refinement iterations rather than over video frames, predicts $u^r_{tj} \in \mathbb{R}^3$, converted to a bounded axis-angle increment,

$$\omega^r_{tj} = \alpha_j \frac{\tanh u^r_{tj}}{\sqrt{1 + \|\tanh u^r_{tj}\|^2}}, \qquad R^{r+1}_{tj} = \exp([\omega^r_{tj}]_\times)\, R^r_{tj}, \tag{4}$$

with $\tanh$ applied componentwise and $\alpha_j = 2°$ for the root and $5°$ otherwise. Since $\|\tanh u\| < \sqrt{3}$ and $s/\sqrt{1+s^2}$ is increasing, each increment is smaller than $\frac{\sqrt{3}}{2}\alpha_j$, *i.e.* $1.73°$ for the root and $4.33°$ for other joints, and the triangle inequality on rotations bounds the displacement after three iterations by $5.2°$ and $13.0°$ relative to the refiner input. Shape and camera pass through unchanged, while projections, local samples and the routed evidence are recomputed after every update.

Objective. All variants use one objective: joint and vertex errors in metres, rotation and shape supervision, velocity and acceleration of the hip-centred prediction-minus-target joint error, a Procrustes-aligned joint auxiliary, intermediate decoder supervision and small consistency and update penalties (coefficients in Supp. S2). Every person-frame is owned by one window, and acceleration triplets that straddle window boundaries are retained by forwarding their owner contexts, so the motion objective and metric cover complete sequences. No 2D reprojection loss is used; external keypoints act only as inputs.

4. Experiments

4.1. Protocol

Data. We use 3DPW [29] with its official splits. The refiner trains on the training split (2,909 eight-frame windows, 22,735 person-frames); every model is evaluated on validation (12 sequences, 10,413 person-frames, 10,226 acceleration triplets) and, once, on test (24 sequences, 35,515 person-frames, 34,752 triplets). A test evaluation is only accepted through an immutable lock that binds the checkpoint hash, configuration, weight type and complete validation result. Person crops follow the dataset's 2D person annotations with a 25% margin, and track identities come from the annotations; Supp. S3 lists these and other deviations from detector-and-tracker pipelines. Earlier models of this project were evaluated on the test split before this study; no decision reported here used test results, but that exposure cannot be undone.

Metrics. MPJPE uses the 14 LSP joints regressed with the H36M regressor; PA-MPJPE additionally applies a per-frame similarity Procrustes alignment. Root conventions differ across the literature, so we report MPJPE and PVE both relative to the H36M pelvis (joint 0; SPIN, CameraHMR, ReFit, PromptHMR) and relative to the hip midpoint (VIBE evaluator, WHAM, GVHMR), and compare each published row only under its own convention. Controlled tables use the pelvis root; our historical unaligned PVE is kept in the supplement. Acceleration error (Accel) is the second difference of the hip-centred prediction-minus-target joint error over all annotation-adjacent triplets, including triplets across inference windows, in mm/frame^2; multiplied by 0.9 it gives m/s^2 at 30 fps, the unit used by WHAM and GVHMR. We apply no flip averaging and no temporal smoothing.

Training and selection. Every variant starts from the same parent and trains for 1,000 steps with identical data order, losses and staged unfreezing (AdamW, effective batch of 16 windows; Supp. S2), with seeds 2026, 2027 and 2028; the router initialisation seed follows the training seed. We evaluate the EMA weights at step 1,000 for every variant and seed, a choice fixed before training; no variant, step or seed is selected on validation or test. The archived parent (P), C10 and C00 checkpoints are evaluated with the same code path.

Statistics. We report the mean and standard deviation over seeds. A paired comparison gives the difference Δ of the frame-weighted aggregate error, averaged over the paired seeds (negative favours the first variant), and a 95% percentile interval from a hierarchical cluster bootstrap [8] that resamples seeds and then sequences (20,000 resamples), because frames within a sequence are strongly correlated. We also count the sequences whose seed-averaged error decreases; paired t-tests over seeds are listed in the supplement but have little power with three seeds. The comparisons in Tab. 3 and the rule that a difference is reliable only if its interval excludes zero and all three seeds agree in sign were fixed before any test evaluation.

4.2. Comparison with published results

Table 1 places our test results next to author-reported numbers, keeping strong variants with additional training data and recording each row's input, training data and evaluation convention. **[TODO: Describe where B4 falls relative to the hip-root video methods (WHAM, GVHMR) and the pelvis-root image methods (CameraHMR, ReFit, PromptHMR) using the matching column only; do not claim state of the art, and recall the annotation-derived crops.]** Temporal quality is a clear weakness of our refiner. The archived C10 checkpoint has an acceleration error of 21.44 mm/frame^2 on validation (archived, single seed), *i.e.* 19.3 m/s^2, whereas WHAM and GVHMR re-

Method	Input; setting	3DPW train	Root	MPJPE↓	PA-MPJPE↓	PVE↓	Accel↓
Author-reported 3DPW test results							
PyMAF [39]	Image; ResNet-50	No	–	92.8	58.9	110.1*	–
PARE [15]	Image; HRNet-W32	No	–	82.0	50.9	97.9*	–
PARE [15]	Image; HRNet-W32	Yes	–	74.5	46.5	88.6*	–
CLIFF [19]	Image; HRNet-W48	Yes	P	69.0	43.0	81.2*	–
PyMAF-X [40]	Image; body-only HR48	No	P	78.0	47.1	91.3*	–
PyMAF-X [40]	Image; body-only HR48	Yes	P	74.2	45.3	87.0*	–
HMR 2.0a [10]	Image; ViT	No	–	70.0	44.5	–	–
HMR 2.0b [10]	Image; ViT	No	–	81.3	54.3	–	–
ReFit [30]	Image; HRNet	Yes	P	65.8	41.0	–	–
ReFit [30]	Image; BEDLAM + real	Yes	P	57.6	38.2	67.6	–
CameraHMR [21]	Image; all data + 3DPW	Yes	P	56.0	35.1	65.9	–
PromptHMR [32]	Image; GT focal length	Yes	P	58.7	36.6	69.4	–
VIBE [14]	Video; SPIN features	Yes	H	82.9	51.9	99.1‡	23.4†
TCMR [5]	Video; temporal context	Yes	H	86.5	52.7	102.9‡	7.1†
GLoT [23]	Video; 16 frames	Yes	H	80.7	50.6	96.3‡	6.6†
WHAM [25]	Video; ViT; real data	Yes	H	57.8	35.9	68.7	6.6
WHAM [25]	Video; ViT; + BEDLAM	Yes	H	56.9	35.7	67.4	6.7
GVHMR [24]	Video; flip evaluation	Yes	H	55.6	36.2	67.2	5.0
PromptHMR [32]	Video; no flip averaging	Yes	P	56.9	35.5	67.3	–
OnlineHMR [43]	Causal online video	Yes	–	69.9	43.7	83.7*	–
Ours, 3DPW test, annotation-derived crops, no flip or smoothing; root P / H							
Parent P	Video; HMR 2.0a + VACE	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**
C10 (archived, routes drift)	+ head routing	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**
A1 mean (C00 design)	+ head mean	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**
B7 ordered vector	+ ViTPose, concatenated	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**
B4 STaR-Body (routed)	+ ViTPose, routed	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**
B6 head + ViTPose	+ both, routed	Yes	P / H	**TBD / TBD**	**TBD**	**TBD / TBD**	**TBD**

Table 1. **3DPW test results** (mm; Accel in m/s^2). Published rows are the authors' own numbers with their inputs and training data. Root gives the convention under which MPJPE and PVE were computed: P, vertices and joints relative to the H36M-regressed pelvis; H, relative to the hip midpoint; –, not reported, or convention or unit not established by our audit of the source. Our rows give both conventions (P / H) and are means over three seeds (single checkpoints for P and C10); compare each published row only with the matching value. *PVE alignment not audited. ‡Unaligned PVE against neutral-gender ground-truth bodies, which our evaluator does not reproduce; not comparable. †Frame-based acceleration as printed (mm/frame^2, not converted). Our crops are derived from dataset annotations, so these rows are not a like-for-like ranking (Sec. 4.1).

port 6.6 and 5.0 m/s^2 on test. Our refiner applies no temporal smoothing and is not a motion prior: temporal attention exchanges features between frames, and the acceleration term is one loss among several over a short continuation. **[TODO: Confirm with TBD m/s^2 and state whether routing changes temporal error (Tab. 3, Accel column).]**

4.3. Does real evidence help, and through which interface?

Table 2 reports the core variants and Tab. 3 their paired differences on test. Continuing the parent without any landmark channel (A0) isolates the effect of the extra budget: A0 reaches **TBD** mm test MPJPE, against **TBD** mm for the parent.

Self-referential evidence. Mean (A1) and routed (A2) head feedback re-run the archived C00/C10 designs with three seeds. Against no feedback, routed drift gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences), and routed against mean drift gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). **[TODO: If both CIs include 0: state that the archived C10 advantage does not survive three seeds and the test split, consistent with routing a signal that contains no image evidence. If A2 beats A0: discuss the reference-drift signal as a learned pull toward the reference (regulariser), not evidence.]**

External evidence. Routing ViTPose evidence instead of drift changes test MPJPE by Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences), and relative to no feedback by Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). **[TODO: State whether real evidence helps (CI of B4 vs A0 excludes 0) and by how much on PVE (Tab. 3).]**

Identity and routing. The central comparison holds the evidence fixed and changes only the interface. Against the capacity-matched pooled control, routing gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences); against ordered-vector conditioning, which keeps identity by position with 3.1× more parameters, it gives

				Validation				Test			
ID	Evidence	Interface	Params	MPJPE	PA-MPJPE	PVE	Accel	MPJPE	PA-MPJPE	PVE	Accel
P	parent, no continuation		–	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
A0	none	–	0	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
A1	head (drift)	mean (C00)	512	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
A2	head (drift)	routed (C10)	105,472	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
B1	ViTPose	pooled	100,096	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
B7	ViTPose	ordered vector	329,344	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
B4	ViTPose	routed (ours)	105,472	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 2. **Controlled comparison** on 3DPW (mean±std over three seeds; mm, Accel in mm/frame^2; pelvis root, pelvis-aligned PVE). All variants continue the same parent P for 1,000 steps and are evaluated with EMA weights at that pre-registered step. "Head (drift)" denotes the inherited observation head, whose outputs equal the frozen reference projection (Sec. 3.2). Params counts the feedback module parameters that receive gradient. Archived single-seed validation values for reference: parent 61.26/37.64/73.02, C10 61.19/37.61/72.82, C00 61.26/37.64/73.05 mm (MPJPE/PA-MPJPE/unaligned PVE; C00 transcribed from run records).

Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). **[TODO: Decision rule (reliable = CI excludes 0 and all seeds agree in sign). B4 better than both B1 and B7: anatomically routed tokens beat pooled and ordered-vector (WHAM-style) conditioning at equal evidence. B4 $\approx$ B7, both better than B1: identity matters but routing is not required; argue parameter efficiency (105k *vs.* 329k), interpretability and multi-source fusion (B6). B4 $\approx$ B1 $\approx$ A0: external evidence does not help this refiner; reframe the paper around the self-referential-evidence diagnostic and the controlled protocol.]**

4.4. What does the router need?

Table 4 varies one ingredient of the router at a time. Removing explicit identity and bias (B2) gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences); removing only the bias (B3) gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences); and a deranged anatomy (B5) gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). **[TODO: Interpret: if B5 is clearly worse than B4 but B3 is not, the bias matters only when wrong (attention learns anatomy on its own); if B3 and B5 are both worse, the anatomical prior helps at this budget; if none differs, identity embeddings carry the effect. Keep the implicit-identity caveat of Sec. 3.4 for B2.]**

Evidence quality and fusion. With the OpenPose detections shipped with 3DPW, routing *vs.* pooling gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences), and ViTPose *vs.* OpenPose under routing gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). Adding the self-referential head tokens to ViTPose (B6) gives Δ=**TBD** mm (95% CI **TBD**; lower on **TBD** sequences). **[TODO: Interpret: does the routing advantage grow with evidence quality (compare B4-vs-B1 with C4-vs-C1)? Does reference drift add anything once real evidence is present (B6 vs B4)? The C runs are optional; drop this paragraph and the C rows if they were not run.]**

4.5. Where does routing act?

Figure 3 breaks the aggregate differences down. Sequence-level changes (Fig. 3a) show whether an aggregate difference is carried by a few sequences or distributed across motions, and the confidence analysis (Fig. 3b) tests the expectation that evidence feedback helps most where the detector is confident and the parent is wrong. **[TODO: Describe the observed pattern: number of sequences improved (TBD for B4 vs B1), sequences with the largest gains or losses, and whether low-confidence frames are protected by the $-2 \log c$ feature.]**

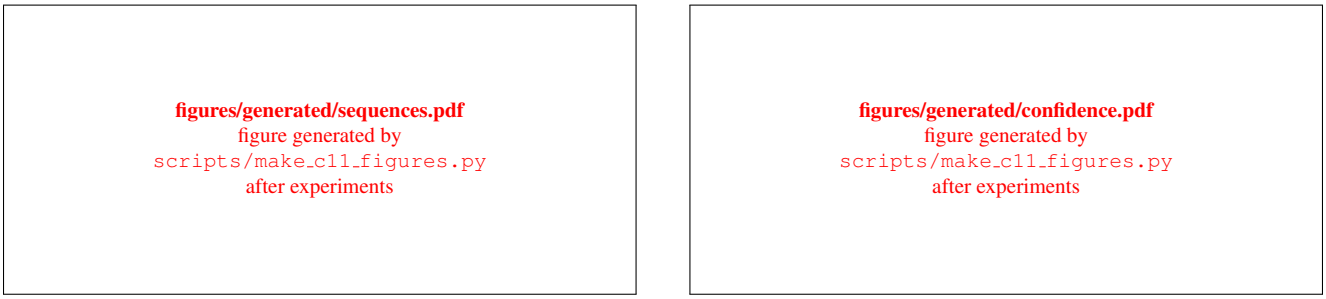
Limitations. Our evaluation uses annotation-derived crops and ground-truth track identity for the shape prototype and the validity masks, so the absolute numbers are likely optimistic relative to a detector-and-tracker pipeline; the paired comparisons share this protocol, but their size may change with detected crops. All conclusions rest on one dataset, three seeds and one continuation budget; the router may matter more or less with longer training or other data, and a world-grounded benchmark such as EMDB [13] remains to be evaluated. The temporal error of our refiner is far above that of motion-prior methods (Sec. 4.2). Finally, ViTPose runs on these annotation-derived crops, so its keypoints may be more accurate than in a full detection pipeline.

5. Conclusion

We asked whether preserving landmark identity and routing it anatomically matters for video mesh refinement once the evidence is real. An audit showed that our earlier joint-indexed feedback routed drift from a frozen reference, so we rebuilt the interface around external keypoints that become identity-bearing tokens, are read by joint queries

Comparison	Isolated factor	ΔMPJPE [95% CI]	Wins	ΔPVE [95% CI]	Wins	ΔPA-MPJPE	ΔAccel
B4 *vs.* B1	routing *vs.* capacity-matched pooling	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* B7	routing *vs.* ordered vector	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* B2	explicit identity + bias	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* B3	anatomical bias	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* B5	correct *vs.* deranged anatomy	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* A2	external *vs.* self-referential evidence	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B6 *vs.* B4	adding head tokens	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B1 *vs.* A1	external *vs.* drift, pooled	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
A2 *vs.* A1	routed *vs.* mean drift (C10 *vs.* C00)	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
A1 *vs.* A0	mean drift *vs.* none	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
A2 *vs.* A0	routed drift *vs.* none	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* A0	routed evidence *vs.* none	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
C4 *vs.* C1	routing *vs.* pooling, OpenPose	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD
B4 *vs.* C4	ViTPose *vs.* OpenPose, routed	TBD TBD	TBD	TBD TBD	TBD	TBD	TBD

Table 3. **Paired differences on 3DPW test** (first minus second variant; mm, Accel in mm/frame^2; negative favours the first). Δ is the frame-weighted aggregate difference averaged over paired seeds; intervals are 95% hierarchical cluster-bootstrap intervals over seeds and sequences; Wins counts the 24 sequences whose seed-averaged error decreases. The comparisons were fixed before test evaluation; validation counterparts and paired t-tests are in Supp. S6.



(a) Per-sequence ΔPVE on test

(b) Effect *vs.* detector confidence

Figure 3. **Where routing acts.** (a) Seed-averaged ΔPVE (pelvis-aligned) of B4 against B1 for the 24 test sequences, sorted; blue bars are improvements. (b) Test MPJPE change of B4 against A0, binned by the mean ViTPose confidence of a person-frame (quintiles; sequence-cluster bootstrap band). All paired comparisons with their intervals are plotted in Supp. S6.

ID	Id.	Bias	Evidence	MPJPE	PA	PVE	Accel
B4	✓	anatomy	ViTPose	TBD	TBD	TBD	TBD
B2	–	–	ViTPose	TBD	TBD	TBD	TBD
B3	✓	–	ViTPose	TBD	TBD	TBD	TBD
B5	✓	deranged	ViTPose	TBD	TBD	TBD	TBD
B6	✓	anatomy	head + ViTP.	TBD	TBD	TBD	TBD
C4	✓	anatomy	OpenPose	TBD	TBD	TBD	TBD
C1	–	pooled	OpenPose	TBD	TBD	TBD	TBD

Table 4. **Router ablations** on 3DPW test (means over three seeds; mm, Accel in mm/frame^2; pelvis root). Id.: learned landmark identity; Bias: kinematic-tree bias in Eq. (3). Standard deviations and validation results are in Supp. S6.

with a kinematic-tree bias, and are recomputed as the body moves. A multi-seed study from a common parent, with test evaluation behind validation locks and sequence-clustered intervals, separates the evidence source, identity, routing, anatomical prior and evidence quality. **[TODO: State the answer in one or two sentences, matching Sec. 4.3: which factor changes test error with a CI excluding zero, and which do not. If routing is not distinguishable from ordered-vector conditioning, conclude that the evidence source, not the interface, drives the gain, and that interface claims require controls of this kind.]**

LLM Usage Statement

Earlier drafts of this work used OpenAI Codex for manuscript editing, cross-checking sources and results, reference-metadata verification and document preparation; Google Gemini supplied an overview reference and OpenAI image generation produced drafts of the overview and landmark-feedback figures, which were then edited and checked against the implemented computation. Human, mesh and attention illustrations in these figures are schematic. In this revision, Anthropic's Claude assisted with the code of the evidence router and controls, the experiment and analysis tooling, and the drafting of this LaTeX manuscript. All reported numbers come from archived measurements, cited original papers, or the released evaluation and analysis scripts, which write them into the manuscript automatically; no language or image model produced

experimental outcomes. The authors verified the code, the analysis and the text and take full responsibility for them.

References

[1] Thomas Bachlechner, Bodhisattwa Prasad Majumder, Henry Mao, Garrison Cottrell, and Julian McAuley. ReZero is all you need: Fast convergence at large depth. In *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*, 2021. 2

[2] Fabien Baradel, Matthieu Armando, Salma Galaaoui, Romain Brégier, Philippe Weinzaepfel, Grégory Rogez, and Thomas Lucas. Multi-HMR: Multi-person whole-body human mesh recovery in a single shot. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024. 2

[3] Michael J. Black, Priyanka Patel, Joachim Tesch, and Jinlong Yang. BEDLAM: A synthetic dataset of bodies exhibiting detailed lifelike animated motion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[4] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. OpenPose: Realtime multi-person 2D pose estimation using part affinity fields. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(1):172–186, 2021. 1, 2

[5] Hongsuk Choi, Gyeongsik Moon, Ju Yong Chang, and Kyoung Mu Lee. Beyond static features for temporally consistent 3D human pose and shape from a video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1964–1973, 2021. 1, 2, 6

[6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[7] Sai Kumar Dwivedi, Yu Sun, Priyanka Patel, Yao Feng, and Michael J. Black. TokenHMR: Advancing human mesh recovery with a tokenized pose representation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1323–1333, 2024. 2

[8] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993. 5

[9] Guénolé Fiche, Simon Leglaive, Xavier Alameda-Pineda, and Francesc Moreno-Noguer. MEGA: Masked generative autoencoder for human mesh recovery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5366–5378, 2025. 2

[10] Shubham Goel, Georgios Pavlakos, Jathushan Rajasegaran, Angjoo Kanazawa, and Jitendra Malik. Humans in 4D: Reconstructing and tracking humans with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14783–14794, 2023. 1, 2, 6

[11] Zeyinzi Jiang, Zhen Han, Chaojie Mao, Jingfeng Zhang, Yulin Pan, and Yu Liu. VACE: All-in-one video creation and editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 17191–17202, 2025. 2

[12] Angjoo Kanazawa, Michael J. Black, David W. Jacobs, and Jitendra Malik. End-to-end recovery of human shape and pose. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7122–7131, 2018. 1, 2

[13] Manuel Kaufmann, Jie Song, Chen Guo, Kaiyue Shen, Tianjian Jiang, Chengcheng Tang, Juan José Zárate, and Otmar Hilliges. EMDB: The electromagnetic database of global 3D human pose and shape in the wild. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 7

[14] Muhammed Kocabas, Nikos Athanasiou, and Michael J. Black. VIBE: Video inference for human body pose and shape estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5253–5263, 2020. 1, 2, 6

[15] Muhammed Kocabas, Chun-Hao P. Huang, Otmar Hilliges, and Michael J. Black. PARE: Part attention regressor for 3D human body estimation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11127–11137, 2021. 2, 6

[16] Nikos Kolotouros, Georgios Pavlakos, Michael J. Black, and Kostas Daniilidis. Learning to reconstruct 3D human pose and shape via model-fitting in the loop. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2252–2261, 2019. 1, 2

[17] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 3744–3753, 2019. 2, 4

[18] Jiefeng Li, Chao Xu, Zhicun Chen, Siyuan Bian, Lixin Yang, and Cewu Lu. HybrIK: A hybrid analytical-neural inverse kinematics solution for 3d human pose and shape estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2

[19] Zhihao Li, Jianzhuang Liu, Zhensong Zhang, Songcen Xu, and Youliang Yan. CLIFF: Carrying location information in full frames into human pose and shape estimation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 590–606, 2022. 2, 6

[20] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Transactions on Graphics*, 34(6): 248:1–248:16, 2015. 1

[21] Priyanka Patel and Michael J. Black. CameraHMR: Aligning people with perspective. In *International Conference on 3D Vision (3DV)*, 2025. 2, 6

[22] István Sárándi and Gerard Pons-Moll. Neural localizer fields for continuous 3D human pose and shape estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2

[23] Xiaolong Shen, Zongxin Yang, Xiaohan Wang, Jianxin Ma, Chang Zhou, and Yi Yang. Global-to-local modeling for

video-based 3D human pose and shape estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8887–8896, 2023. 1, 2, 6

[24] Zehong Shen, Huaijin Pi, Yan Xia, Zhi Cen, Sida Peng, Zechen Hu, Hujun Bao, Ruizhen Hu, and Xiaowei Zhou. World-grounded human motion recovery via gravity-view coordinates. In *SIGGRAPH Asia Conference Papers*, pages 144:1–144:11, 2024. 1, 2, 6

[25] Soyong Shin, Juyong Kim, Eni Halilaj, and Michael J. Black. WHAM: Reconstructing world-grounded humans with accurate 3D motion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2070–2080, 2024. 1, 2, 6

[26] Anastasis Stathopoulos, Ligong Han, and Dimitris Metaxas. Score-guided diffusion for 3D human recovery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 906–915, 2024. 2

[27] Team Wan et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. 2

[28] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017. 2

[29] Timo von Marcard, Roberto Henschel, Michael J. Black, Bodo Rosenhahn, and Gerard Pons-Moll. Recovering accurate 3D human pose in the wild using IMUs and a moving camera. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 601–617, 2018. 2, 5

[30] Yufu Wang and Kostas Daniilidis. ReFit: Recurrent fitting network for 3D human recovery. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14644–14654, 2023. 2, 6

[31] Yufu Wang, Ziyun Wang, Lingjie Liu, and Kostas Daniilidis. TRAM: Global trajectory and motion of 3D humans from in-the-wild videos. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 467–487, 2024. 2

[32] Yufu Wang, Yu Sun, Priyanka Patel, Kostas Daniilidis, Michael J. Black, and Muhammed Kocabas. PromptHMR: Promptable human mesh recovery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1148–1159, 2025. 2, 6

[33] Yan Xia, Xiaowei Zhou, Etienne Vouga, Qixing Huang, and Georgios Pavlakos. Reconstructing humans with a biomechanically accurate skeleton. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5355–5365, 2025. 2

[34] Yufei Xu, Jing Zhang, Qiming Zhang, and Dacheng Tao. ViTPose: Simple vision transformer baselines for human pose estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 1, 3

[35] Yufei Xu, Jing Zhang, Qiming Zhang, and Dacheng Tao. ViTPose++: Vision transformer for generic body pose estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(2):1212–1230, 2024. 1, 3

[36] Chengxuan Ying, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, 2021. 2

[37] Ye Yuan, Umar Iqbal, Pavlo Molchanov, Kris Kitani, and Jan Kautz. GLAMR: Global occlusion-aware human mesh recovery with dynamic cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11038–11049, 2022. 2

[38] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 2, 4

[39] Hongwen Zhang, Yating Tian, Xinchi Zhou, Wanli Ouyang, Yebin Liu, Limin Wang, and Zhenan Sun. PyMAF: 3D human pose and shape regression with pyramidal mesh alignment feedback loop. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11446–11456, 2021. 2, 6

[40] Hongwen Zhang, Yating Tian, Yuxiang Zhang, Mengcheng Li, Liang An, Zhenan Sun, and Yebin Liu. PyMAF-X: Towards well-aligned full-body model regression from monocular images. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10):12287–12303, 2023. 2, 6

[41] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2

[42] Yuhong Zhang, Guanlin Wu, Ling-Hao Chen, Zhuokai Zhao, Jing Lin, Xiaoke Jiang, Jiamin Wu, Zhuoheng Li, Hao Frank Yang, Haoqian Wang, and Lei Zhang. HumanMM: Global human motion recovery from multi-shot videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1973–1983, 2025. 2

[43] Yiwen Zhao, Ce Zheng, Yufu Wang, Hsueh-Han Daniel Yang, Liting Wen, and László A. Jeni. OnlineHMR: Video-based online world-grounded human mesh recovery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13951–13961, 2026. 2, 6

[44] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5745–5753, 2019. 2